*CME 108/MATH 114*
*Introduction to Scientific Computing*
*Summer 2020*

# Rootfinding
# Bisection method

**Guiding question** How can I solve an equation? In particular, given a function $f$, how can I find $x$ such that $f(x) = 0$?

**Definition 1.** A *zero* or a *root* of $f$ is an element $x$ in the domain of $f$ such that $f(x) = 0$.

*Remark* 1. Note that the problem of solving an equation $f(x) = a$ reduces to finding a root of $g(x) = f(x) - a$. In the general case both $x$ and $f$ are vector-valued (systems of equations) but for now we'll focus on scalar-valued functions of a single var.

The most interesting case in our course is made un by functions which cannot be solved for analytically.

**Example 2.** *(Vander Waals equations) Recall from introductory chemistry the ideal gas law $PV = nRT$ used to model ideal gases. Real gases are in fact not fully compressible and there are attractive forces among their molecules, so a better model for their behavior is*

$$\big(P + \frac{n^2 a}{V^2}\big)\big(V - nb\big) = nRT, \tag{1}$$

*where $a$ and $b$ are correction terms. In a lab, suppose 1 mol of chlorine gas has a pressure of 2 atm and a temperature of $313K$, for chlorine, $a = 6.29$ atm $L^2/mol^2, b = 0.0562L/mol$. What is the volume?*

*We can't just "isolate" $V$. (Granted, in this simple case we obtain a low-degree polynomial in $V$ and there are special methods for finding their roots. In this special case, the cubic formula will suffice.)*

**Bisection Method** (Enclosure vs fixed point iteration schemes).

A basic example of enclosure methods: knowing $f$ has a root $p$ in $[a, b]$, we "trap" $p$ in smaller and smaller intervals by halving the current interval at each step and choosing the half containing $p$.

Our method for determining which half of the current interval contains the root $p$ is based on the *intermediate value theorem.*

**Theorem 3** (IVT)**.** *Let $f$ be a continuous function on $[a, b]$ and let $k$ be any number between $f(a)$ and $f(b)$. Then there exists $c$ in $(a, b)$ such that $f(c) = k$.*

Informally, "A continuous function on an interval achieves all values between its values at the end points."

What does the IVT tell us about root finding?

Consider $\operatorname{sgn}(f(a)f(b))$. If $f(a)f(b) < 0$, what can you say?

*Remark* 2. Note that the same conclusion holds regardless of the magnitude of $f(a)$ and $f(b)$: only the sign of their product matters.

**Example 4.** $f(x) = x^3 + 2x^2 - 3x = 1$

$$\begin{aligned} f(-3) &= -1, \quad & f(-1) &= 3, \quad & f(1) &= -1 \\ f(-2) &= 5, \quad & f(0) &= -1, \quad & f(2) &= 9. \end{aligned}$$

*Draw a picture! See Figure 1.*

**The Method** Begin with an interval $[a, b]$ such that $f(a) \cdot f(b) < 0$. Find $p = (a + b)/2$. Test wether $f(a) \cdot f(p) < 0$. If so, then $f$ has a root in $[a, p]$. Make $[a, p]$ the new interval and repeat the process. If not, then $f(a)$ and $f(p)$ have the same sign and therefore we are guaranteed that $f(p)f(b) < 0$ (since $f(a)$ and $f(b)$ have opposite signs), which then implies $f$ has a root in $[p, b]$.

Make this the new interval and repeat the process. Generate a sequence of interval $[a_n, b_n]$ each guaranteed to contain a root of $f$ and at each step use the approximation

$$p_n = \frac{a_n + b_n}{2} \tag{2}$$

of the enclosed root. Stop when $p_n$ is "close enough" to a root of $f$ in $p$.

The following example might suggest how to devise a stopping condition.

**Example 5.** *Square root via bisection.*

$f(x) = x^2 - 2$, *start with* $[0, 2]$

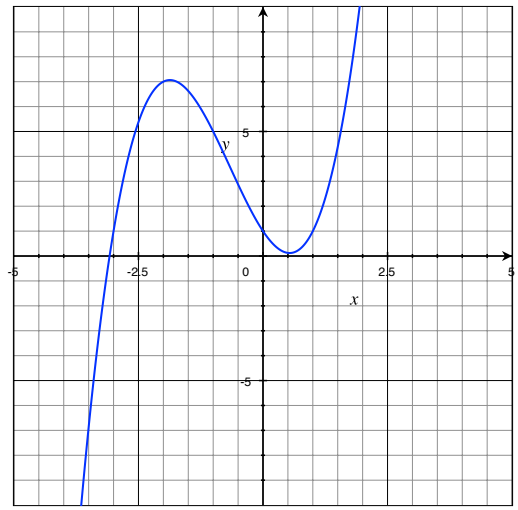


Figure 1: $f(x) = x^3 + 2x^2 - 3x - 1$

1. *Note* $f(0) \cdot f(2) = (-2)(2) = -4 < 0$, *so* $f$ *indeed has a root in* $[0, 2]$ *by IVT. Set* $p_1 = 1$ *and notice* $f(0) \cdot f(1) = (-2)(-1) > 0$, *so choose* $[a_2, b_2] = [1, 2]$.

2. *Set* $p_2 = \frac{1+2}{2} = \frac{3}{2}$ *and notice* $f(a_2) \cdot f(p_2) = (-1)(\frac{9}{4} - 2) = \frac{-1}{4} < 0$ *so choose* $[a_3, b_3][1, \frac{3}{2}]$.

3. *Set* $p_3 = \frac{1+\frac{3}{2}}{2} = \frac{5}{4}$ *and notice* $f(a_3) \cdot f(p_3) = (-1)(\frac{25}{16} - 2) = \frac{7}{32} > 0$, *so choose* $[a_4, b_4] = [\frac{5}{4}, \frac{3}{2}]$.

4. *Now* $p_4 = \frac{\frac{5}{4}+\frac{3}{2}}{2} = \frac{11}{8} = 1.375\ldots$

Note that at iteration $n$ our root approximation $p_n$ is contained in an interval $[a_n, b_n]$ that is half the size of the previous interval $[a_{n-1}, b_{n-1}]$. Are we guaranteed convergence? If so in which cases?

**Theorem 6.** *Let* $f$ *be a continuous function on* $[a, b]$ *and suppose that* $f(a) \cdot f(b) < 0$. *Then the bisection method generates a sequence of iterates* $p_n$ *which converges to a root* $p \in (a, b)$ *with the property that*

$$|p_n - p| \leq \frac{b - a}{2^n}. \tag{3}$$

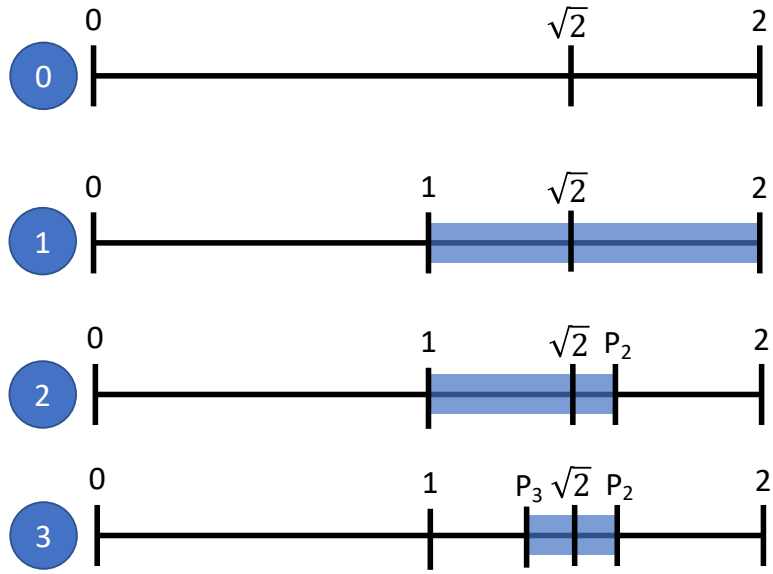


Figure 2: Bracketing $f(x) = x^2 - 2$ on $[0, 2]$

*Remark* 3.

1. Note conclusion: $p_n$ converges to *a* root of $f$. If there are multiple roots in $(a, b)$, we can't a priori know to which one $p_n$ will converge.

2. We get a theoretical error bound, $|p_n - p|$, which can be helpful to eradicate bugs when coding.

3. Method cannot be used for locating roots of *even multiplicity*.

**Definition 7.** A root $p$ of the equation $f(x) = 0$ is said to be of *multiplicity* $m$ if $f$ can be written as $f(x) = (x - p)^m q(x)$, with $\lim_{x \to p} q(x) \neq 0$.

Equivalently, $p$ is of multiplicity $m$ if $f(p) = f'(p) = \cdots = f^{(m-1)}(p) = 0$ and $f^{(m)}(p) \neq 0$ (assuming $f$ is a smooth function).

We'll say more about multiplicity in the coming lectures.

*Proof.* We need only establish the error bound to prove convergence, since $\frac{b-a}{2^n} \to 0$ as $n \to \infty$.

By construction of the method, at each step we are guaranteed $p \in (a_n, b_n)$, and $p_n$ is the midpoint of $(a_n, b_n)$. This implies

$$|p_n - p| \leq \frac{b_n - a_n}{2}. \tag{4}$$

However,

$$\begin{aligned} b_n - a_n &= \frac{1}{2}(b_{n-1} - a_{n-1}) \\ &= \frac{1}{2} \cdot \frac{1}{2}(b_{n-2} - a_{n-2}) \\ &\cdots \\ &= \frac{1}{2^{n-1}}(b_1, a_1). \end{aligned}$$

□

The convergence theorem suggests the stopping criterion $(b_n - a_n)/2 < \epsilon$. Since the absolute error $|p_n - p|$ is guaranteed to be no larger than $(b_n - a_n)/2$, this stopping criterion guarantees the root approximation $p$ is no further than $\epsilon$ from $p$. That is, if we stop iterating when

$$\frac{b_n - a_n}{2} < \epsilon,$$

we are guaranteed that $|p_n - p| < \epsilon$.

Thus we obtain the following procedure.

**Pseudocode**

**Algorithm 1** Bisection Method

```
Given f, [a, b], ϵ, N_max
sfa ← sign(f(a))
for i ← 1 to N_max do
    p ← (a + b)/2
    if (b − a)/2 < ϵ then
        return p
    end if
    sfp ← sign(f(p))
    if sfa · sfp < 0 then
        b ← p
    else
        a ← p
        sfa ← sfp
    end if
end for
```